

# Where the constraint migrates in biotech: finite atoms and human donors that capital cannot replicate

*Six calls on inputs that scale with physics or human arms, not with money: isotope purity, legacy radium, enriched stable targets, donated plasma, donated platelets, and the organ pool GLP-1 quietly drains.*

**Thesis spine:** Frontier → Capability → Dependency graph → Supply elasticity → Demand → Capital → Pricing → Policy → Outcomes. Rent accrues to the inelastic input. The edge is naming where the constraint moves before pricing catches up.

Area: biotech and human health (drug modalities and delivery, biomanufacturing and fill-finish capacity, donor-derived and cell/gene inputs, diagnostics and tools/reagents, medical isotopes and devices, longevity and aging) • Horizon: 2030 to 2040

Method: generate wide and disruptive, then gate strict. Each call names the needle, not the theme.

Two probabilities per call: directional vision, and the strict dated clause scored at resolution.

Status: hardened candidates (survived the adversarial refute pass). Drafted 2026-06-14.

## The board: 6 hardened calls

### The cross-cutting read

Every one of these six calls rests on the same structural move: a loud boom is funding the visible layer (curie count, fractionation plants, targeting ligands, reconditioning devices) while the binding constraint sits one layer upstream on an input that is physically or biologically non-manufacturable on the relevant timescale. Three are medical-isotope calls, but they name three genuinely different inelastic needles and must not be collapsed. The Ac-227 call is about isotopic inseparability: Ac-225 and Ac-227 are the same element in the same oxidation state, so every scaled accelerator route co-produces a radiotoxic impurity that no chemistry removes, and only the finite Th-229/U-233 generator stock is clean by construction. The Ra-226 call is about quantity: roughly 2.5 kg of legacy radium was ever extracted, the IAEA "fix" is a bounded gleaning of antique hospital sources rather than a manufacturing ramp, and building accelerators only intensifies competition for that fixed stock. The stable-isotope call is about geography: kilogram-scale gas-centrifuge enrichment of Yb-176, Zn-68 and Gd-160 (the targets that breed Lu-177, Cu-67, Tb-161) sits overwhelmingly in Rosatom cascades, and Western EMIS/laser rebuilds are gram-scale and isotope-narrow. The two donor-derived calls share the deepest physical truth: polyclonal IgG and platelet-lysate growth factors are not molecules you synthesize, they are pooled human repertoires whose therapeutic property is donor diversity, so supply is hard-bounded by eligible donors times a regulated max donation rate. The GLP-1 call inverts the consensus sign: by removing the metabolic and cardiovascular deaths that feed the marginal donor stream, GLP-1 degrades rather than improves the usable organ pool. Across all six, the pre-consensus seam is consistent and the clause probabilities are honestly held below the vision probabilities for one recurring reason: these constraints clear through government allocation tables, bilateral contracts, and informal hospital rationing rather than through Bloomberg-trackable spot prices, so the structural call can be correct while the dated, publicly-cited resolution criterion fails to print on schedule.

### At a glance

| #  | The boom                                                                                                                                                                                                                                                                                                                                                          | Binding constraint (the needle)                                                                                          | Vision P | Clause P | Resolves   |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------|----------|------------|
| P1 | Targeted alpha therapy: Ac-225-PSMA for prostate cancer, Ac-225-DOTATATE for neuroendocrine tumors, and the broad wave of antibody- and small-molecule-targeted alpha conjugates. Market analysts project 20 to 44 percent CAGR toward 1 to 4 billion dollars by 2030, with at least nine commercial alpha products expected by 2030. Pharma majors are racing to | Isotopically pure, Ac-227-free Ac-225 sourced from the finite, non-replenishing Th-229/U-233 stockpile and the handful o | 74%      | 57%      | 2032-12-31 |

| #  | The boom                                                                                                                                                                                                                                                                                                                                                                                                                                              | Binding constraint (the needle)                                                                                          | Vision P | Clause P | Resolves   |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------|----------|------------|
|    | lock supply (Lilly/Point, Novartis, BMS/RayzeBio, AstraZeneca, Bayer). The current narrative treats raw Ac-225 curie availability as the bottleneck and the targeting vector as the innovation layer.                                                                                                                                                                                                                                                 |                                                                                                                          |          |          |            |
| P2 | An Ac-225-based targeted alpha therapy reaches blockbuster volume. The field treats Ac-225 atoms and accelerator beam-time as the things to scale, while the non-replenishable Ra-226 target feedstock quietly becomes the true ceiling.                                                                                                                                                                                                              | Purified Ra-226 target feedstock (roughly 2.5 kg total global legacy stock, non-manufacturable, recoverable only from an | 78%      | 44%      | 2034-12-31 |
| P3 | Pluvicto and Lutathera turned beta-emitting radioligand therapy into a multi-billion-dollar oncology category, and capital is now flooding into the headline layer: targeting ligands, reactor slots, cyclotrons, and the radioisotopes themselves (Tb-161, Cu-67, Ac-225). The reactor chokepoint is the public narrative.                                                                                                                           | Kilogram-scale gas-centrifuge enrichment of stable-isotope targets (Yb-176, Zn-68, Gd-160) concentrated in Rosatom plant | 72%      | 38%      | 2032-12-31 |
| P4 | Polyclonal IgG is the pooled antibody repertoire of thousands of donors. Unlike any monoclonal or recombinant biologic, it cannot be grown in a cell line because the therapeutic property is the diversity of the pooled-donor immune repertoire, not a single sequence. That biological fact is permanent and non-negotiable. Demand is being pulled upward by three compounding forces: aging populations with primary immunodeficiency, expanding | Human source-plasma collection litres from the US-concentrated, compensated donor base -- the irreplaceable polyclonal-I | 82%      | 40%      | 2035-12-31 |

| #  | The boom                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Binding constraint (the needle)                                                                                                 | Vision P | Clause P | Resolves   |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------|----------|------------|
|    | <p>approved neurology indications (CIDP, MG, multifocal motor neuropathy), and a growing population of cancer-immunotherapy survivors with secondary immunodeficiency. IG demand is tracking roughly 30 percent growth by 2030 and rising into the 2030s. Supply is litres of source plasma, 70 percent of which is collected from US compensated donors under regulatory and physiological frequency caps that set a hard ceiling on litres per donor per year. Each lot requires 7-12 months from collection to finished vial. The system cannot sprint. The rent concentrates in whoever controls donor-centre litres, and that input cannot be financialized, engineered, or scaled the way a factory can.</p> |                                                                                                                                 |          |          |            |
| P5 | <p>The allogeneic off-the-shelf cell therapy and regenerative-medicine wave: MSC products, iPSC-derived therapies, allogeneic CAR-T and NK products, and the broader stem-cell pipeline that must expand cells to billions of doses in bioreactors. Regulatory and quality pressure is simultaneously forcing the field off FBS, which is itself supply-capped and carries zoonotic risk, onto a human-derived substitute whose supply is biologically constrained and whose donor pool is already strained for ordinary clinical transfusion.</p>                                                                                                                                                                 | <p>GMP-grade human platelet lysate derived from pooled volunteer-donor blood-bank platelet units. The inelastic input is hu</p> | 72%      | 38%      | 2033-12-31 |

| #  | The boom                                                                                                                                                                                                                                                                                                                                                                                                        | Binding constraint (the needle)                                                                                                | Vision P | Clause P | Resolves   |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------|----------|------------|
| P6 | <p>The loud GLP-1 narrative is about supply and demand in obesity/diabetes therapeutics and downstream food/device/CDMO effects. The transplant-adjacent GLP-1 conversation is entirely about GLP-1 as therapy for recipients and for reversing donor-organ steatosis. Everyone treats GLP-1 as straightforwardly good for organ health. Nobody models GLP-1 as a structural drag on the donor pool itself.</p> | <p>Standard-criteria-donor (SCD) organs: young, low-comorbidity deceased-donor solid organs usable without reconditioning.</p> | 68%      | 38%      | 2033-12-31 |

Vision P = strength of the structural case. Clause P = calibrated odds the exact dated clause resolves true, scored with Brier. The gap is the honest timing and measurement tax, not timidity.

## P1 · Isotopically pure (Ac-227-free) Ac-225 from the finite Th-229/U-233 stockpile is the physics-locked binding constraint on targeted-alpha therapy through the 2030s, not raw curie count

P1

The boom: Targeted alpha therapy: Ac-225-PSMA for prostate cancer, Ac-225-DOTATATE for neuroendocrine tumors, and the broad wave of antibody- and small-molecule-targeted alpha conjugates. Market analysts project 20 to 44 percent CAGR toward 1 to 4 billion dollars by 2030, with at least nine commercial alpha products expected by 2030. Pharma majors are racing to lock supply (Lilly/Point, Novartis, BMS/RayzeBio, AstraZeneca, Bayer). The current narrative treats raw Ac-225 curie availability as the bottleneck and the targeting vector as the innovation layer. · Domain: biotech and human health (medical isotopes / radiopharmaceuticals)

DIRECTIONAL VISION

**74%**

STRICT CLAUSE

**57%**

RESOLVES

**2032-12-31**

Alpha emitters kill tumors by delivering high-LET, double-strand DNA breaks over a few cell diameters. There is no chemical or biological substitute: the physics of the decay is the drug. Historically, all clinical-grade Ac-225 came from Th-229 generators isolated from an aged U-233 stockpile built for weapons and reactor programs. Th-229 decays to Ac-225 with no Ac-227 branch, so generator-derived material is isotopically clean. Legacy U.S. supply at ORNL is on the order of 1 Curie per year. U-233 production is permanently shut down under current policy, making this source finite and non-replenishing. Every scaled accelerator route under development (proton or deuteron irradiation of Ra-226 targets, spallation of Th targets, photonuclear routes targeting more than 100 Ci/yr by 2029) co-produces Ac-227 because the nuclear reactions are adjacent. Ac-225 and Ac-227 are the same element in the same oxidation state: no chemical separation is possible. Ac-227 has a 21.8-year half-life, adding persistent radiotoxic burden to the patient and waste stream. The Ac-227 ceiling on cumulative patient dose tightens as TAT regimens move from single doses to repeat or chronic administration, the trajectory being driven by emerging indications. NorthStar's April 2026 FDA DMF acceptance for "no-carrier-added" Ac-225 from a cyclotron route signals the industry is aware of purity as a quality attribute, but the distinction between carrier-free chemistry purity and Ac-227 isotopic purity has not yet been resolved in public regulatory language or sell-side models. RayzeBio's Phase 3 pause over Ac-225 shortage is the early tremor showing that supply gating is real; the purity dimension will emerge as the next layer as curies scale but clean curies do not.

### WHY IT IS PRE-CONSENSUS

Consensus has loudly noticed raw Ac-225 scarcity and is pouring capital into curie expansion (Cardinal, NorthStar, TerraPower, PanTera, SHINE, Eckert and Ziegler). That first-order shortage is priced into pharma supply agreements and equity models. What is not priced is the second-order, physics-locked distinction: scaled routes co-produce Ac-227 inseparably, so adding curies does not add clean curies. Sell-side models and equity coverage treat Ac-225 supply as a single fungible number without separating generator-grade from accelerator-grade or pricing the Ac-227 ceiling on cumulative patient dose. The NorthStar April 2026 DMF acceptance uses "no-carrier-added" language that conflates chemistry purity with isotopic purity, suggesting even regulators and suppliers have not yet formally distinguished the two. The purity premium is forming but is not yet visible in contract pricing or drug master file specifications in the public record. This is a genuine seam not covered in existing FUTURE\_MAP biotech calls, none of which address medical isotope chokepoints.

#### HONEST PRICE CHANNEL

Not priced at the isotopic-purity layer. Raw Ac-225 scarcity is reflected in pharma supply agreements and private market valuations for isotope producers. The Ac-227 purity distinction does not yet appear in public sell-side models, contract pricing disclosures, or regulatory guidance distinguishing generator-grade from accelerator-grade for repeat-dosing indications. The NorthStar NCA DMF language is the closest public signal, but it does not resolve the Ac-227 versus carrier-free ambiguity. Price channel is partially obscure on the needle as specified.

#### THE NEEDLE

Isotopically pure, Ac-227-free Ac-225 sourced from the finite, non-replenishing Th-229/U-233 stockpile and the handful of generators derived from it. The inelastic input is not gross curie output but isotopic purity: the absence of the chemically inseparable Ac-227 co-impurity that physics dictates every scaled accelerator and reactor route introduces.

#### LEADING METRIC

Two tracked series, both annual from 2026. (1) Curies per year of isotopically pure, generator-sourced (Th-229/U-233) Ac-225 with certified Ac-227 content below the dose-acceptable threshold, versus total Ac-225 curies produced including accelerator/Ra-226/photonuclear routes. Current generator-grade supply is approximately 1 Ci/yr at ORNL while accelerator targets aim for 100-plus Ci/yr by 2029. (2) The price and contract-allocation spread per millicurie between certified-low-Ac-227 lots and accelerator-route lots, plus the count of TAT clinical or commercial programs that specify a maximum Ac-227 impurity concentration in their drug master file or supply contract. Resolution criterion: does the clean-grade share remain a small minority of total supply while commanding a rising purity premium, and does Ac-227 impurity become an explicitly cited gating specification in TAT supply agreements by 2032?

#### KILL-CRITERION

Kill if by 2032 an accelerator/reactor/photonuclear route reaches routine multi-tens-of-Curie annual output of Ac-225 whose Ac-227 content is certified low enough that regulators and sponsors treat it as interchangeable with generator-grade for chronic or repeat dosing (purity premium collapses to under 15 percent and no major TAT program cites Ac-227 as a supply or dosing constraint). Also kill if practical electromagnetic or mass-selective isotope separation removes the Ac-227 impurity at commercial scale, or if the field migrates decisively to Pb-212 or Tb-149 alpha emitters that sidestep the Ac-225 purity problem. Partial kill: if U-233 production is reinstated under a policy reversal, the finite-stockpile premise weakens even if the purity advantage of Th-229-derived material stays intact.

#### REFUTE CHECK (SURVIVED)

Three attacks survive at reduced severity. First, single-dose or low-repeat TAT protocols (the current standard) may never accumulate Ac-227 to a regulatory ceiling, keeping the purity constraint theoretical unless chronic dosing regimens scale -- the call's load-bearing assumption. Second, U-233 production closure is policy, not physics: a future nuclear materials program could reopen the route, weakening the finite-stockpile premise. Third, Pb-212 alpha therapy (from Ra-228/Th-228 generators) is a live alternative modality that sidesteps the Ac-227 problem entirely; if it captures a large share of the TAT pipeline, the purity constraint becomes niche. None of these attacks kill the call today, but they set the real odds below the structural ceiling because they are plausible within the resolution window.

**Why this call earned a place** The physics of co-production is locked: Ac-225 and Ac-227 are the same element, and every accelerator scaling route produces both. The Th-229/U-233 source is the only route that avoids this by construction, and that source is permanently capped under current policy. The purity premium is not yet priced into equity models or supply contracts. The call survives all three adversarial attacks at the structural level. Clause probability is held below vision probability because the resolution criterion requires observable market signals (purity premiums in contracts, Ac-227 specs in DMFs) that are forming but not yet locked, and because the chronic-dosing proliferation timeline is uncertain within the 2026 to 2032 window.



## **P2 · By 2034, the binding constraint on the Ac-225 targeted-alpha-therapy boom is not the accelerator conversion step but the finite, non-manufacturable Ra-226 target feedstock that every accelerator route consumes, and the hot-cell capacity to fabricate, irradiate, and recover it.**

The boom: An Ac-225-based targeted alpha therapy reaches blockbuster volume. The field treats Ac-225 atoms and accelerator beam-time as the things to scale, while the non-replenishable Ra-226 target feedstock quietly becomes the true ceiling. · Domain: biotech

DIRECTIONAL VISION

**78%**

STRICT CLAUSE

**44%**

RESOLVES

**2034-12-31**

Ac-225 production has three serial layers: Ra-226 source, accelerator conversion, and GMP radiochemistry. The funded buildout (Niowave, Actineer/CNL cyclotron already running in 2025, DOE photonuclear programs) attacks layer 2. But Ra-226 is a secular decay product of uranium, never made on purpose: roughly 2.5 kg was extracted worldwide across the 20th century, and the only new supply is scavenging antique radium medical sources under the IAEA global radium-management initiative. Accelerator routes do not consume Ra-226 atoms permanently if the target is recycled, but imperfect recovery plus limited hot-cell fabrication capacity means the effective Ra-226 inventory shrinks with throughput. As conversion capacity is funded and built, the binding constraint deterministically walks upstream to the fixed Ra-226 stock and the shielded hot-cell capacity to work that stock without loss. This is structurally identical to the He-3 case: a tiny, byproduct-only legacy inventory that capital cannot replicate on a decade timescale.

### WHY IT IS PRE-CONSENSUS

The investor and trade narrative has moved to "Ac-225 is scarce, fund accelerators and hot-cells." IAEA, DOE/NIDC, and the trade press (AuntMinnie, Clinical Trial Vanguard) already name Ra-226 sourcing as a feedstock concern, so the framing is not buried. However, it has not crossed into sell-side equity models, where accelerator capacity remains the consensus binding variable. The second-order migration (solving conversion just exposes the Ra-226 floor) is absent from financial modeling. Pre-consensus position is real but eroding: awareness is rising in technical and institutional channels. The price-channel leg (no tracked spot price for Ra-226) is the cleanest indicator that financial markets have not priced this yet.

### HONEST PRICE CHANNEL

No public spot or contract price for purified Ra-226 exists. DOE/NIDC sells research quantities under administrative allocation, not open-market bidding. This means the structural constraint is real but price discovery may remain inside government allocation tables rather than a Bloomberg-trackable index, which is the primary risk to clause resolution independent of whether the structural call is correct.

### THE NEEDLE

Purified Ra-226 target feedstock (roughly 2.5 kg total global legacy stock, non-manufacturable, recoverable only from antique radium sources) plus the GMP shielded hot-cell capacity to fabricate, irradiate, and chemically recover those targets without radium loss.

### LEADING METRIC

Public citations by an Ac-225 sponsor or CDMO of Ra-226 target availability as the rate-limiting input; emergence of a tracked spot or contract price for purified Ra-226 (\$/mg or \$/Ci); reported Ra-226 inventory recovered under the IAEA radium initiative versus annual Ac-225 demand. Baseline: global Ac-225 roughly 1.7 Ci/yr (enough for about 2000 patients); no public Ra-226 price index; DOE/NIDC lists Ra-226 as available in research quantities only; accelerator-conversion buildout is the entire headline story in equity and trade press.

#### KILL-CRITERION

The call is dead if, by 2034, accelerator buildout makes Ac-225 broadly available with no sponsor or CDMO citing Ra-226 target feedstock or target-fab hot-cell capacity as rate-limiting and no Ra-226 price or inventory tension emerges; OR if a Ra-226-free route to Ac-225 (high-energy proton spallation of Th-232 at TRIUMF or DOE scale, or Ra-226-free reactor routes) scales enough to make Ra-226 inventory irrelevant to supply; OR if Ac-225 alpha therapies fail pivotal efficacy or toxicity readouts and Lu-177 beta therapy remains dominant at scale.

#### REFUTE CHECK (SURVIVED)

Three adversarial attacks: (1) Ra-226-free routes. High-energy proton spallation of Th-232 at TRIUMF and BNL produces Ac-225 without Ra-226 targets. If these scale commercially before 2034, Ra-226 becomes irrelevant. Real risk, but multi-GeV proton accelerators are expensive and sparse; commercial displacement of Ra-226 routes by 2034 is not current consensus. (2) Already named publicly. NIDC, IAEA, and AuntMinnie already cite Ra-226 sourcing. If a sell-side analyst picks this up soon, the "not priced" premise weakens fast. The resolution clause asks for a tracked spot price, which requires open-market price discovery; government allocation could prevent this even as the constraint bites. (3) Target recycling loop. Accelerator routes can recycle Ra-226 after irradiation if hot-cell recovery is efficient. The constraint then shifts to recovery yield and hot-cell capacity rather than Ra-226 atoms outright. The needle should name both Ra-226 inventory and hot-cell recovery capacity together, which the candidate already does. The structural logic survives all three attacks, but the exact clause resolution is vulnerable to the government-allocation pricing mechanism.

**Why this call earned a place PROMOTE** on structural strength: the Ra-226 supply inelasticity is physically real, the constraint-migration logic from conversion to feedstock is deterministic as accelerator capacity builds, and the comparison to He-3 is apt. The pre-consensus gap is genuine in financial channels. Clause probability is held to 0.44 because the resolution clause requires a tracked spot or contract price for Ra-226, which may never emerge if DOE continues administrative allocation rather than open-market bidding, and because Ra-226-free spallation routes are a live kill-criterion risk. The structural vision (Ra-226 becomes the real ceiling) has a high probability of being correct; the dated clause test is tighter than the structural story alone.

### **P3 · The binding constraint on the RLT boom migrates off the reactor onto kilogram-scale enriched stable-isotope feedstock (Yb-176, Zn-68, Gd-160) concentrated in Russian gas-centrifuge cascades, with Western rebuilds too gram-scale and too isotope-narrow to close the gap across the full therapeutic basket by 2032.**

The boom: Pluvicto and Lutathera turned beta-emitting radioligand therapy into a multi-billion-dollar oncology category, and capital is now flooding into the headline layer: targeting ligands, reactor slots, cyclotrons, and the radioisotopes themselves (Tb-161, Cu-67, Ac-225). The reactor chokepoint is the public narrative. · Domain: biotech-health / medical isotopes / radiopharmaceuticals / stable-isotope enrichment

DIRECTIONAL VISION

**72%**

STRICT CLAUSE

**38%**

RESOLVES

**2032-12-31**

A therapeutic radioisotope is bred, not mined. No-carrier-added Lu-177 requires an ytterbium-176-enriched target bombarded in a high-flux reactor; Cu-67 needs zinc-68; Tb-161 needs gadolinium-160. Enriching a stable isotope to greater than 98 percent at kilogram scale requires gas-centrifuge cascades, and that capacity sits overwhelmingly at Rosatom's Electrochemical Plant (Zelenogorsk) and Ural Electrochemical (Novouralsk). The reactor or cyclotron only activates the target; it cannot produce the feedstock. Western centrifuge capacity for stable isotopes was retired in the 1990s. The rebuild is underway via EMIS (Kinectrics, ORNL SIPRC) and laser enrichment (ASP Isotopes ASPI, Pretoria), but Kinectrics EMIS tops out near 500 g/yr for Yb-176 and ASP Isotopes has not disclosed sustained kilogram-per-year throughput at clinical grade. Zn-68 and Gd-160 have no announced kilogram-scale Western rebuild programs. As RLT scales 5-10x and diversifies across isotopes, the rent step-shifts to whoever controls enriched-stable-target supply, one layer above the reactor, and the West has identified the dependency but has not solved it at basket scale. The constraint is isotope-specific and slow to qualify under GMP, making it genuinely inelastic through 2030 even if capital is now allocated.

#### **WHY IT IS PRE-CONSENSUS**

The first-order dependency (Yb-176 from Russia for Lu-177) is now in trade press, EU consortium papers (EURASIS 2024), and equity narratives (ASPI listed on Nasdaq explicitly to trade this thesis). That layer is partially priced. The second-order claim is less priced: that the constraint is not the reactor, not Lu-177 specifically, and not Yb-176 alone, but the upstream gas-centrifuge enrichment step shared across the entire next generation of therapeutic isotopes as a basket, with Western rebuilds solving one isotope partially while leaving Zn-68 and Gd-160 unaddressed, and with allocation contracts keeping the bottleneck invisible to spot-market signals and equity coverage. Policy and equity narratives price the drug and the reactor while overlooking the centrifuge two layers upstream. Western EMIS announcements (Kinectrics 500 g/yr) are treated as a solution when the demand arithmetic does not support that conclusion at full basket scale. Distinct from existing biotech calls in FUTURE\_MAP (delivery, AAV, plasmid, ADC conjugation), none of which touch medical isotopes or stable-isotope enrichment.

#### **HONEST PRICE CHANNEL**

Partially priced at the first-order level via ASPI equity and DOE SIPRC funding. The second-order basket argument (Zn-68, Gd-160 remaining unaddressed; enriched-target allocation as the shared constraint

across all next-generation therapeutic isotopes) is not reflected in any visible spot price or dedicated equity vehicle. Enriched target prices are set by bilateral allocation contracts without public disclosure, making the constraint structurally invisible to markets even when it is operationally real.

#### THE NEEDLE

Kilogram-scale gas-centrifuge enrichment of stable-isotope targets (Yb-176, Zn-68, Gd-160) concentrated in Rosatom plants; Western EMIS and laser rebuilds are gram-scale for Yb-176 and nonexistent at scale for the rest of the therapeutic basket.

#### LEADING METRIC

Share of Western RLT therapeutic-isotope feedstock (enriched Yb-176, Zn-68, Gd-160) sourced from non-Russian kilogram-scale enrichment, plus disclosed Yb-176 unit price and lead time. Mid-2026 baseline: NCA Lu-177 supply chain still Russia-dependent for Yb-176 at commercial scale; Kinectrics targets 500 g/yr Yb-176; ASP Isotopes has produced commercial samples but no disclosed kg/yr throughput; no Western kilogram program announced for Zn-68 or Gd-160; enriched-target lead times measured in months and allocated by quiet bilateral contract rather than spot market.

#### KILL-CRITERION

Western or allied centrifuge-equivalent stable-isotope enrichment reaches qualified kilogram-scale output for at least two of {Yb-176, Zn-68, Gd-160} before 2030 (e.g. ASP Isotopes MLIS scales to greater than 2 kg/yr at clinical grade for Yb-176 AND a second target isotope enters a kilogram-scale Western program), OR accelerator-direct production routes that bypass enriched stable targets scale enough to supply commercial RLT without target enrichment, OR RLT demand growth stalls such that enriched-target allocation pressure is not rate-limiting for any program.

#### REFUTE CHECK (SURVIVED)

Three attacks survive partial scrutiny. First, the resolve clause requires public citation by RLT sponsors of enriched-target supply as the rate-limiting input over reactor time; CDMOs and sponsors routinely describe supply chain risk in vague 10-K language, and if allocation is secured bilaterally, sponsors have no incentive to disclose the specific bottleneck -- making the evidentiary bar hard to meet even if the bottleneck is real. Second, the Kinectrics plus ASPI plus SIPRC overlap could solve Yb-176 specifically by 2029, splitting the clause: Western capacity may exceed demand for Lu-177 while remaining inadequate for Tb-161 and Cu-67, producing an ambiguous resolve rather than a clean TRUE. Third, six years is a long runway -- the West has identified the dependency and capital is moving; by 2032 the basket may be partially addressed for the leading commercial isotopes even if not fully closed. The structural observation survives all three attacks; the specific dated clause is the weaker component.

**Why this call earned a place** Promoted because the second-order structural mechanism is correct and inelastic through the mid-term, the basket argument (Zn-68, Gd-160 have no Western kilogram rebuild) extends beyond what equity or policy has priced, and the allocation-contract invisibility is a genuine market-structure reason the bottleneck does not self-disclose. clause\_p is held below 0.5 because the evidentiary bar (public citation by sponsors of enriched target as rate-limiting input over reactor time, for two isotopes) may not be met on schedule even if the bottleneck is operationally real, and because ASP Isotopes plus Kinectrics could solve Yb-176 before 2032, splitting the resolve.

**P4 · By 2035 the binding constraint on IVIG/SCIG supply is the human source-plasma donor base, a demographically and geographically concentrated input with no recombinant substitute; structural rationing or allocation tightens in at least two major wealthy markets, and the US share of world plasma collection stays above ~65 percent even as global IG demand grows materially.**

The boom: Polyclonal IgG is the pooled antibody repertoire of thousands of donors. Unlike any monoclonal or recombinant biologic, it cannot be grown in a cell line because the therapeutic property is the diversity of the pooled-donor immune repertoire, not a single sequence. That biological fact is permanent and non-negotiable. Demand is being pulled upward by three compounding forces: aging populations with primary immunodeficiency, expanding approved neurology indications (CIDP, MG, multifocal motor neuropathy), and a growing population of cancer-immunotherapy survivors with secondary immunodeficiency. IG demand is tracking roughly 30 percent growth by 2030 and rising into the 2030s. Supply is litres of source plasma, 70 percent of which is collected from US compensated donors under regulatory and physiological frequency caps that set a hard ceiling on litres per donor per year. Each lot requires 7-12 months from collection to finished vial. The system cannot sprint. The rent concentrates in whoever controls donor-centre litres, and that input cannot be financialized, engineered, or scaled the way a factory can. · Domain: biotech-health / donor-derived biologics (plasma fractionation)

DIRECTIONAL VISION

**82%**

STRICT CLAUSE

**40%**

RESOLVES

**2035-12-31**

Source plasma is collected from humans at a near-fixed litres-per-donor-per-year ceiling set by regulation and physiology. The donor pool is concentrated in one country and skewed toward a compensated demographic. There is no recombinant escape for polyclonal IgG because the therapeutic property IS the pooled-donor repertoire diversity. Demand is driven by aging plus neurology indication expansion plus immunocompromised-survivor population growth; supply is driven by human arms and a 7-12 month fractionation pipeline. The curves are structurally decoupled and the input is non-substitutable and demographically capped.

#### WHY IT IS PRE-CONSENSUS

Partially pre-consensus. The biology of non-substitutability is a permanent fact and is not contested. The shortage risk, however, is more priced than the candidate implies. Grifols, CSL Behring, and Takeda are all publicly traded or have public debt; sell-side analysts at Jefferies, Morgan Stanley, and Berenberg have covered plasma collection volume as the primary operating driver for at least five years. The "US supplies ~70 percent of world plasma" figure appears in WHO working group documents and regulatory filings. What remains genuinely underpriced: the convergence of three demand curves simultaneously (neurology expansion, immunocompromised-survivor population, aging-primary-immunodeficiency), the specific demographic fragility of compensated donation under any economic-normalisation scenario, and the non-availability of any FcRn-class substitute for the primary-immunodeficiency (non-neurology) IVIG base. The claim that this is entirely un-modeled is too strong; the claim that the full binding severity is priced is also too strong. The truth is in between, which warrants PROMOTE but with clause\_p well below vision\_p.

#### HONEST PRICE CHANNEL

Plasma fractionator equities trade on collection-volume growth as the primary lever; the shortage risk is a standing topic on earnings calls. Market research projections (USD 13B to USD 23B by 2035, CAGR ~5.9

percent) are widely cited. FcRn antagonist launches (efgartigimod, rozanolixizumab) are already partially priced as a neurology-demand offset. The US-geographic concentration is in public regulatory filings. This is not a dark, uncovered corner of the market.

#### THE NEEDLE

Human source-plasma collection litres from the US-concentrated, compensated donor base -- the irreplaceable polyclonal-IgG feedstock. Not fractionation plant capacity, not albumin or Factor co-products, not any recombinant alternative (which cannot exist for polyclonal IgG).

#### LEADING METRIC

US share of global plasma collection (currently near 70 percent); IG demand growth vs collected source-plasma litre growth (demand tracking roughly +30 percent by 2030, collection per capita roughly flat); number of major wealthy markets (US, EU member states, Canada, Japan) imposing IVIG allocation, prioritization criteria, or formal rationing. Current state: chronic localized IVIG shortages recurring, non-self-sufficient nations dependent on US plasma, a 2025 PMC study documenting 9 percent therapy-delay rates from shortages, and donor management flagged as a demographic risk in regulatory filings.

#### KILL-CRITERION

By 2035 either (a) FcRn antagonists (efgartigimod class) or other recombinant immunomodulators substitute for enough IVIG demand in the neurology indications that pooled-plasma IgG stops being the binding modality for the fastest-growing demand cohort, OR (b) plasma collection broadens enough via EU/Asia self-sufficiency programs or higher per-donor yield that collected litres outpace IG demand, US share of collection falls below 60 percent, and no major market resorts to IVIG allocation. Either dissolves the donor-base constraint as the binding needle.

#### REFUTE CHECK (SURVIVED)

Three attacks. First, FcRn antagonists are not hypothetical: argenx's efgartigimod (Vyvgart) is commercially approved for MG and CIDP and growing, competing directly for the highest-dose IVIG users in neurology -- the exact demand cohort the candidate calls open-ended. Over a 9-year horizon, this class is a material kill path, not a speculative one. Second, EU plasma self-sufficiency is an active policy target: the European Commission's Plasma Action Plan (2023) and post-COVID health-security legislation have made non-US plasma collection a genuine political priority with capital behind it. Nine years is enough time for this to move the US share below 65 percent. Third, the rationing clause is fragile: formal public rationing in two major wealthy markets requires both governments and hospital systems to move from informal shortage management (the current state) to declared allocation frameworks, which has not happened despite years of recurring shortages, suggesting the system absorbs the stress through informal means rather than triggering the formal resolution criterion. The biological core of the call survives all three attacks; the specific dated clause does not survive them cleanly.

**Why this call earned a place PROMOTE** because the biological mechanism is real, the needle (donated human plasma litres) is genuinely inelastic and non-substitutable for the primary-immunodeficiency base, demand is compounding from three independent curves, and the call is directionally correct with a 9-year horizon. Downgrade clause\_p materially below vision\_p to reflect the three active kill paths: FcRn substitution in neurology, EU self-sufficiency policy, and the fragility of the formal-rationing

resolution criterion. The structural foresight is sound; the exact clause as written has a live 60 percent probability of failing on its specific terms even as the underlying constraint tightens.



## **P5 · Donor-derived human platelet lysate is the hidden, non-synthesizable feedstock that becomes the binding constraint on industrial-scale allogeneic cell therapy as the field migrates off FBS onto a supplement that can only be made from human blood donors -- a donor pool that is already chronically short for clinical transfusion.**

The boom: The allogeneic off-the-shelf cell therapy and regenerative-medicine wave: MSC products, iPSC-derived therapies, allogeneic CAR-T and NK products, and the broader stem-cell pipeline that must expand cells to billions of doses in bioreactors. Regulatory and quality pressure is simultaneously forcing the field off FBS, which is itself supply-capped and carries zoonotic risk, onto a human-derived substitute whose supply is biologically constrained and whose donor pool is already strained for ordinary clinical transfusion. · Domain: biotech and human health (cell/gene therapy upstream inputs, donor-derived biologics)

DIRECTIONAL VISION

**72%**

STRICT CLAUSE

**38%**

RESOLVES

**2033-12-31**

Cells in culture require a serum supplement to proliferate. FBS is supply-capped and carries prion/viral/zoonotic risk that regulators increasingly reject. The leading xeno-free replacement, hPL, is made by pooling and lysing human platelet concentrates from volunteer donors. There is no synthetic route: the growth-factor cargo (PDGF, TGF-beta, IGF, EGF, FGF variants) is biologically inseparable from the human platelet, and pooling 40-80 donor units per production batch is required to average donor-to-donor variability. This hard-couples the cell-therapy industry's input supply to the human blood-donation pool. That pool is physically inelastic: it scales only with willing donors and apheresis-chair capacity, already runs seasonal and structural shortages for clinical transfusion, and has a hard biological ceiling. As allogeneic programs scale from thousands to millions of doses in liter-to-hundreds-of-liter bioreactors at 5% supplement, hPL demand competes with hospital transfusion services for the same platelet units. Recombinant growth-factor cocktails and chemically defined media exist but have not achieved broad GMP validation across MSC, iPSC-derived, and allogeneic NK cell types; regulatory requalification for process changes adds 2-5 years per product. Substitution is real but incomplete, and allogeneic scale-up is outrunning defined-media validation today.

### **WHY IT IS PRE-CONSENSUS**

Market-research reports (Technavio, Polaris, TowardsHealthcare) name the hPL market and project 15% CAGR growth, so the category is tracked. However, the framing throughout is growth-opportunity-in-a-niche, not supply-constraint-story. No equity analyst covers a public hPL pure-play. No forward price curve exists for hPL. No blood-bank or regulatory body has yet issued allocation guidance balancing transfusion versus manufacturing demand. The field frames hPL as the better, safer FBS replacement -- not as a donor-derived bottleneck that cannot scale past human donation. The donor-platelet-diversion framing is absent from regulatory filings and blood-establishment policy. That is a genuine framing gap: the constraint is pre-consensus at the level that matters for pricing.

### **HONEST PRICE CHANNEL**



No spot-price index for hPL exists. Catalog list prices from STEMCELL Technologies and MP Biomedicals are publicly visible but do not reflect GMP commercial-scale contract pricing, which is negotiated privately. No sell-side coverage of a hPL pure-play. Market-report coverage names the sector without pricing the constraint. The absence of a liquid price channel is consistent with the call being pre-consensus, not evidence it is already priced.

#### THE NEEDLE

GMP-grade human platelet lysate derived from pooled volunteer-donor blood-bank platelet units. The inelastic input is human-donor platelet mass itself (apheresis and whole-blood-derived platelets), which is biologically capped, already short for clinical transfusion, and cannot be synthesized or farmed.

#### LEADING METRIC

Tracked annually from 2026: (1) Estimated liters of GMP hPL consumed by cell-therapy and regenerative-medicine manufacturing versus total national platelet-unit collection, and the share of collected or expired platelet units diverted from transfusion into hPL production. (2) Real price per liter of GMP, pathogen-reduced, fibrinogen-depleted hPL and its trend, plus the count of cell-therapy sponsors disclosing hPL or human-platelet supply as a named supply-chain risk or qualifying second sources in regulatory filings or investor disclosures. (3) Whether blood establishments or regulatory bodies issue allocation guidance balancing transfusion versus manufacturing demand for platelets. Resolution requires real-terms hPL price rise plus at least three clinical-stage or commercial allogeneic programs publicly citing human-platelet or hPL supply as a rate-limiting or dual-sourced input.

#### KILL-CRITERION

Kill if, by 2033, a chemically defined or recombinant serum-free medium achieves broad GMP adoption across MSC, iPSC-derived, and allogeneic NK processes such that hPL is no longer the dominant supplement (major new allogeneic approvals run on defined media and hPL demand growth flattens). Also kill if scalable non-donor sources of the platelet growth-factor cargo emerge at cost-competitive GMP scale (iPSC-derived platelets, bioreactor-grown megakaryocytes, or recombinant factor cocktails), decoupling cell-therapy input supply from the human blood-donor pool. Also kill if hPL real price stays flat and no allogeneic sponsor cites human-platelet supply as a constraint through 2033.

#### REFUTE CHECK (SURVIVED)

Three attacks. First: the category is already in market reports, so it is priced. Weakens on inspection -- reports price the growth opportunity in hPL sales, not the scarcity of donor platelet mass as a bottleneck for cell-therapy producers. These are different claims pointing in opposite directions commercially. Second: defined media and recombinant supplements will substitute before the constraint bites. Non-trivial and the most serious attack. Miltenyi TexMACS and others are active, and some MSC runs are already hPL-free. But no single defined medium matches hPL performance across the full cell-type space today, regulatory requalification adds years per product, and allogeneic scale-up is moving faster than defined-media validation. The kill criterion correctly captures this exit. Third: the volume math does not close by 2033. Possible -- commercial allogeneic products at million-dose scale are still 5-10 years from dominance, and the diversion math may not become visible until post-2033. This attack does not kill the structural case but does compress clause probability. Call survives all three with reduced but positive conviction. Clause probability held at 0.38 because the resolution bar (public price rise plus three named-input disclosures) is specific, the timeline is tight relative to allogeneic scale-up, and defined-media progress is real.

**Why this call earned a place** Promote because the biological mechanism is genuine and non-synthesizable, the inelastic needle is precisely identified (donor platelet mass), the field is systematically mispricing the dependency by framing hPL as a quality upgrade rather than a constrained input, and no liquid price channel or equity coverage currently reflects the donor-diversion risk. The kill criterion is honest and testable. Clause probability is held to 0.38 -- not inflated -- because defined-media substitution is a real competing path and the resolution bar requires public evidentiary disclosure that sponsors have structural incentives to avoid.

## **P6 · GLP-1 mass adoption structurally degrades the deceased-donor organ supply: by 2033 the binding constraint in solid-organ transplantation becomes the vanishing pool of young, low-comorbidity SCD organs, forcing value to migrate onto organ reconditioning (normothermic machine perfusion) and xeno/engineered organ infrastructure rather than donor recruitment.**

The boom: The loud GLP-1 narrative is about supply and demand in obesity/diabetes therapeutics and downstream food/device/CDMO effects. The transplant-adjacent GLP-1 conversation is entirely about GLP-1 as therapy for recipients and for reversing donor-organ steatosis. Everyone treats GLP-1 as straightforwardly good for organ health. Nobody models GLP-1 as a structural drag on the donor pool itself. · Domain: biotech-health (transplantation / organ supply, GLP-1 second-order)

DIRECTIONAL VISION

**68%**

STRICT CLAUSE

**38%**

RESOLVES

**2033-12-31**

The deceased-donor pool is a byproduct of how people die young and healthy. The overdose epidemic supplied roughly 42% of US deceased-donor growth between 2009 and 2019 because overdose victims are young with intact organs. That stream is collapsing: overdose-donor share fell from 16.7% in 2022 to 10.5% by Q1 2025 as overdose mortality dropped roughly 27%. Replacement donors are structurally worse -- post-inflection data show median donor age up, diabetes +24%, hypertension +19%, and rising reliance on donation-after-circulatory-death (DCD) with measurably lower graft survival. GLP-1 mass adoption compounds this in a direction nobody models: it durably reduces obesity, diabetes, cardiovascular and cerebrovascular mortality at population scale, which are exactly the death mechanisms feeding the remaining marginal-donor stream, while doing nothing to replenish the young/traumatic-death SCD stream. Over a decade the deceased-donor pool becomes older, smaller relative to demand, and increasingly composed of organs that need reconditioning to be usable. The forced response is normothermic machine perfusion to recondition marginal/DCD/steatotic organs, plus structural pull toward xeno and engineered organs as the only elastic supply.

### **WHY IT IS PRE-CONSENSUS**

The transplant field discusses the aging/comorbid donor-pool shift and the overdose-donor decline as separate phenomena, and separately celebrates GLP-1 as beneficial for organ health and recipients. The specific second-order synthesis -- that GLP-1 mass adoption removes the metabolic/cardiovascular-death donor stream and thereby tightens the supply of usable young SCD organs, repricing reconditioning and engineered-organ infrastructure as the binding capacity -- appears in zero published transplant or investor analysis. Sell-side covers TransMedics as a technology/logistics adoption story, not as a GLP-1 second-order consequence. The consensus has the sign of GLP-1's effect on transplantation backwards at the population-supply level. Note: the downstream asset (NMP/reconditioning equity) is already partially priced -- TMDX is a high-momentum public compounder -- but the specific causal attribution via GLP-1 donor-pool degradation is genuinely absent from coverage, making this a structural-mechanism insight rather than an undiscovered equity trade.

### **HONEST PRICE CHANNEL**

TransMedics (TMDX) at \$605M 2025 revenue, 36% growth, active Buy-rated analyst coverage at \$109 price target, stock up 81% in 2025. NMP market reports project \$2B (2026) to \$6.5B+ (2033) at 18% CAGR. The reconditioning value migration is therefore partially priced as a technology-adoption story. The GLP-1 donor-pool-degradation causal path is not in any sell-side model or transplant analysis. Verdict: the downstream consequence is partially priced on technology grounds; the specific mechanism named here is not priced. A structural call on the mechanism and its compounding is still pre-consensus; a simple "buy NMP stocks" trade is not.

#### THE NEEDLE

Standard-criteria-donor (SCD) organs: young, low-comorbidity deceased-donor solid organs usable without reconditioning. A pool fed by traumatic/overdose/cerebrovascular young death, now physically shrinking as those death mechanisms recede and GLP-1 removes the metabolic-death replacement stream.

#### LEADING METRIC

(a) SCD share of the US deceased-donor pool and median donor age (OPTN/SRTR annual data); (b) fraction of transplanted organs requiring machine perfusion or reconditioning before implant; (c) DCD and extended-criteria-donor share of total transplants; (d) installed normothermic-machine-perfusion device base and per-organ reconditioning revenue. Current anchors: overdose-donor share 16.7% to 10.5% (2022 to Q1 2025); TransMedics (TMDX) revenue \$605M in 2025, growing 36% year-over-year, active Buy coverage at \$109 target; NMP market projected \$2B in 2026 to \$6.5B by 2033 at 18% CAGR. GLP-1 contribution to donor-pool degradation not yet separated in any published donor-pool analysis.

#### KILL-CRITERION

By 2033, the deceased-donor SCD pool is stable or growing in absolute terms and median donor age is flat-to-falling (a new young-death source or expanded living/altruistic donation offsets the decline), AND machine-perfusion reconditioning has not become a standard step for a rising majority of transplanted organs. OR scaled xeno/engineered organs become the dominant marginal-supply solution, meaning SCD scarcity was bypassed rather than binding. OR GLP-1 population effects demonstrably fail to reduce the comorbid-death donor stream within the window (donor mortality mechanisms unchanged by 2030), falsifying the causal mechanism.

#### REFUTE CHECK (SURVIVED)

Three serious attacks survive review. First, the downstream asset is already priced: TMDX at high-multiple, high-momentum Buy-rated status means the market has partly re-rated NMP as the organ-quality solution. The "value migrates to reconditioning" clause is thus partially already in prices, weakening the investment edge. Second, the GLP-1 mortality timeline is slow: mass adoption began around 2023, cardiovascular/metabolic death-rate effects at population scale take 5-10 years, so by 2033 GLP-1 may have only 10 years at scale and the donor-pool signal may not be separable from other variance within the window. Third, the kill criteria are plausible: DCD expansion is already accelerating as a partial offset, living donation is growing, and xeno-organ trials (FDA-authorized pig-to-human, 2024) are on a plausible timeline to reach limited commercial scale by 2030-2033, potentially bypassing rather than confirming the SCD-scarcity binding constraint. The structural direction is sound; the exact clause -- SCD scarcity becoming THE binding constraint and forcing observable value migration by 2033 -- is weaker because value migration is already underway and the GLP-1 causal contribution is not separable within the window. The call survives as a structural insight but not at high precision on the exact resolution clause.

Why this call earned a place PROMOTE on the structural mechanism: the GLP-1-as-donor-pool-degrader second-order insight is genuinely absent from transplant and investor analysis, the SCD organ needle is genuinely inelastic, and the directional trend (worse donor pool, rising NMP adoption) is already confirmed in real data. Clause\_p held below 0.5 because the downstream equity is partially priced on technology grounds already, the GLP-1 causal contribution is not separable within the 2033 window, and alternative elastic responses (DCD expansion, xeno) may prevent SCD scarcity from becoming definitively "binding" in a measurable sense. This is a real structural insight worth tracking, not a high-confidence binary resolution.

## Seeds considered and not promoted

Cleared the physical-constraint test but failed on investability or on the price channel. Logged because the discipline is to surface what was cut.

| Seed                                                                                                                                                                                                                             | Physical case                                                                          | Why not promoted                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| By 2035, donor-rate ceiling forces formal indication-based IVIG rationing in at least one high-income system, with donor-collection-hours (not fractionation plant capacity) cited as the limiter.                               | Polyclonal immunoglobulin donor-rate ceiling / standing indication-based rationing     | Near-duplicate of P4 (same plasma-donor needle and same FcRn/EU-self-sufficiency kill paths); P4 is the stronger statement of the identical mechanism with higher vision <sub>p</sub> (0.82 vs 0.70) and a cleaner US-concentration metric, so this one is folded in to avoid double-counting the plasma thesis.                                       |
| By 2034, purified Ra-226 legacy feedstock plus licensed Ac-227-free hot-cell separation capacity is the gating variable; at least three pivotal-stage programs publicly ration dose citing isotope feedstock or hot-cell limits. | Ra-226 feedstock + hot-cell separation capacity, three-program public-rationing clause | Substantially overlaps P2 (Ra-226 finite legacy stock) and P1 (Ac-227-free purity) combined into one needle; its clause is more fragile because it requires three separate companies to publicly disclose rationing, which sponsors are structurally incentivized to suppress. P1 and P2 already carry the two distinct physical needles more cleanly. |
| By 2032, Ra-226 target feedstock (not Ac-225 production capacity) becomes the explicitly cited gating variable for the radioligand pipeline.                                                                                     | Ra-226 as explicitly-cited gating variable by 2032                                     | Direct near-duplicate of P2 with the same 2.5 kg legacy-stock needle and IAEA gleaning argument, but a tighter, harder-to-meet clause (requires explicit public Ra-226 citation by 2032 rather than by 2034). P2 captures the identical mechanism with more headroom on both timing and the price/inventory resolution channel.                        |

Generated by the Pope System. Each call is a forward instrument: resolution date and kill-criterion fixed at creation, superseded never edited, clause probability scored with Brier at resolution.